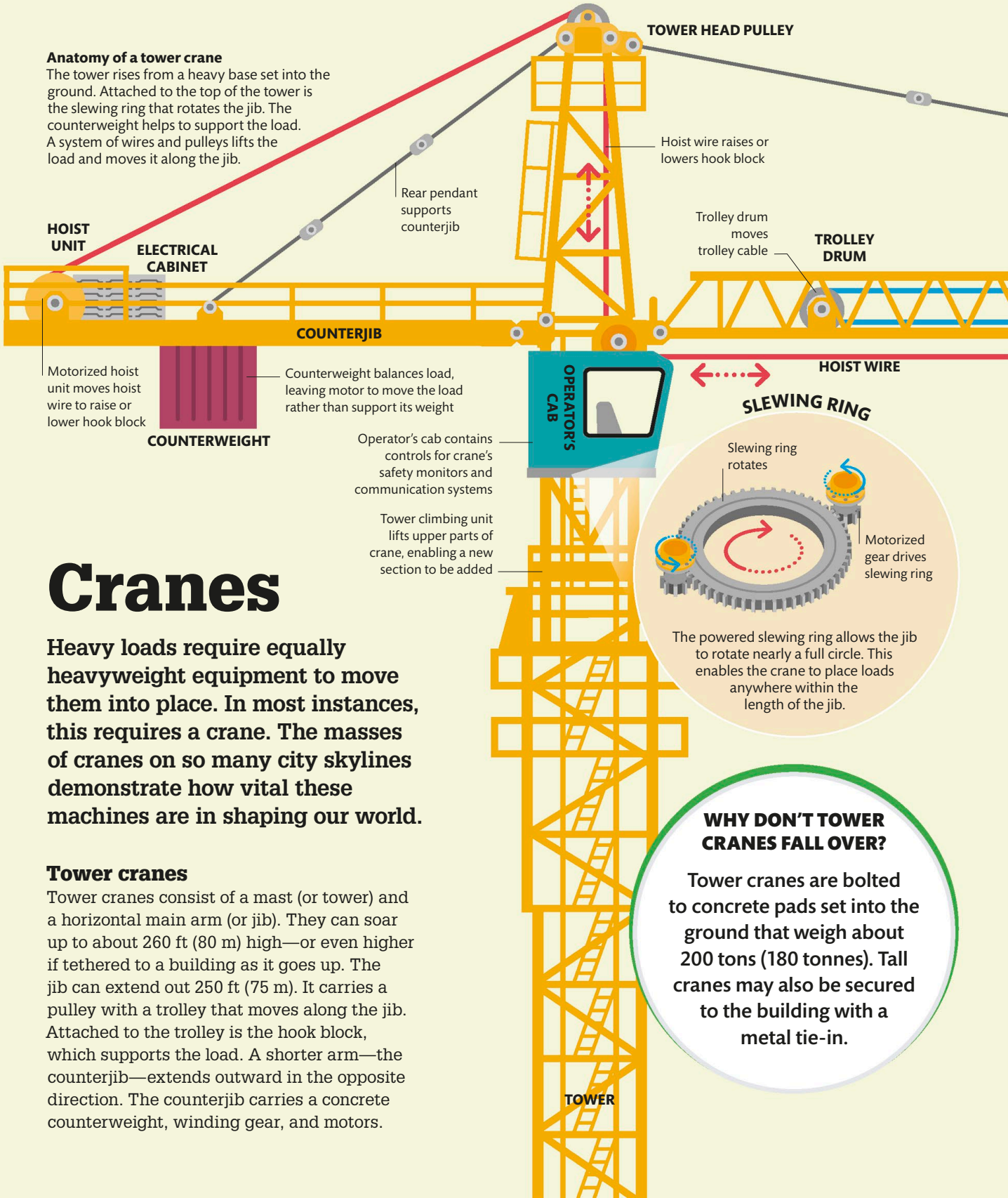
Cranes

Heavy loads require equally heavyweight equipment to move them into place. In most instances, this requires a crane. The masses of cranes on so many city skylines demonstrate how vital these machines are in shaping our world.

Tower cranes

Tower cranes consist of a mast (or tower) and a horizontal main arm (or jib). They can soar up to about 260 ft (80 m) high—or even higher if tethered to a building as it goes up. The jib can extend out 250 ft (75 m). It carries a pulley with a trolley that moves along the jib. Attached to the trolley is the hook block, which supports the load. A shorter arm—the counterjib—extends outward in the opposite direction. The counterjib carries a concrete counterweight, winding gear, and motors.



WHY DON'T TOWER CRANES FALL OVER?

Tower cranes are bolted to concrete pads set into the ground that weigh about 200 tons (180 tonnes). Tall cranes may also be secured to the building with a metal tie-in.